MATH 1342 – Elementary Statistics – Week 13: Inference for Two Means and Paired Data

Nov 17 – 19 | 6.4 Inference for a Difference in Means (Distribution, CI, Hypothesis Test) • 6.5 Paired Difference in Means | Fall 2026 · Section 036 · TTh 6:00–7:20 PM (DIL)

Module Objectives	Instructional Materials	Learning Activities	Assessments	Tools
1. Compute the standard error and conservative degrees of freedom for a difference in two independent means, and check the conditions. (CLO 5) **2.** Construct and interpret a confidence interval for $\mu_1 - \mu_2$, explaining how the order of the groups affects the sign. (CLO 6) **3.** Carry out a two-sample t test, choosing the tail from the research question. (CLO 8)	e-Text: 6.4 Inference for a Difference in Means (6.4-D, 6.4-CI, 6.4-HT) Interactive Lecture: 6.4 Inference for a Difference in Means Slides: 6.4 Inference for a Difference in Means	Participation: Comparing Two Means – video games and GPA CI; Pygmalion-effect test	Online Homework: 6.4 Inference for a Difference in Means (due Sun Nov 22, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
4. Classify a study design as paired or independent samples by checking for a link between observations. (CLO 1, CLO 8) **5.** Define the difference variable d, carry out a paired t test and paired t confidence interval, and explain why pairing reduces variability. (CLO 6, CLO 8)	e-Text: 6.5 Paired Difference in Means Interactive Lecture: 6.5 Paired Difference in Means Slides: 6.5 Paired Difference in Means	Participation: Paired or Independent? – class discussion of eight study scenarios (a)–(h); define d and the order of subtraction	Online Homework: 6.5 Paired Difference in Means (due Sun Nov 22, 11:59 PM)	WileyPLUS StatKey Blackboard Zoom (live class)
Items that cover objectives from more than one row above (objectives in parentheses)			Test 3 – Fri Nov 20 – Sun Nov 22 (Blackboard, proctored, 2 hours). Covers Weeks 10–13. (all objectives)	Assessment Tool in Blackboard Respondus LockDown Browser and Monitor

CLO 1 Explain the use of data collection and statistics as tools to reach reasonable conclusions. **CLO 5** Examine, analyze, and compare various sampling distributions for both discrete and continuous random variables. **CLO 6** Describe and compute confidence intervals. **CLO 8** Perform hypothesis testing using statistical methods.